

Guess, Check, Correct

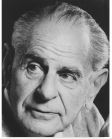
Weekend 1: Vapnik's statistical learning theory

Carlos Cotrini

CAS Building ML/AI Applications, ETH Zurich

Friday 4 and Saturday 5 September 2026

Popper, 1934: the tetradic schema



Karl Popper

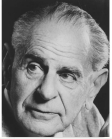
We do not know: we can only guess.

Karl Popper, *The Logic of Scientific Discovery*, chapter 10, section 85 (1959).

Schema after Karl Popper, *Objective Knowledge*, chapter 8 (1972).

Weekend 1, Friday 08:00

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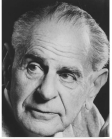
P_1

Problem

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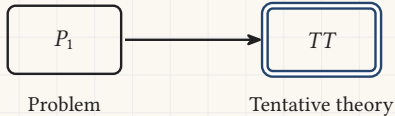
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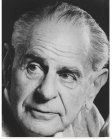
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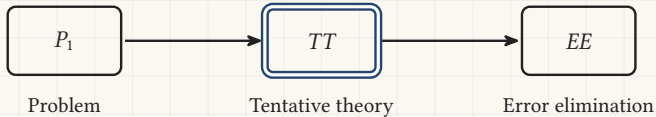
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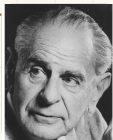
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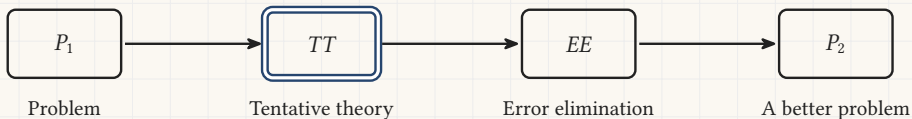
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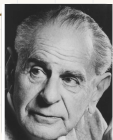
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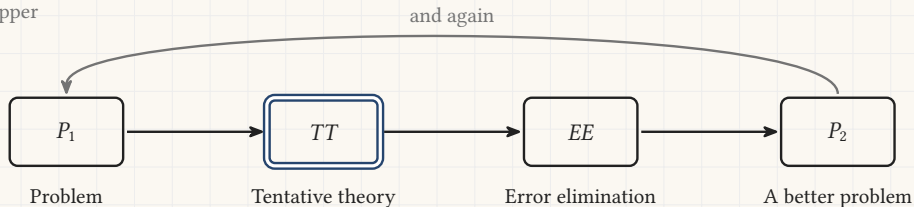
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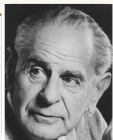
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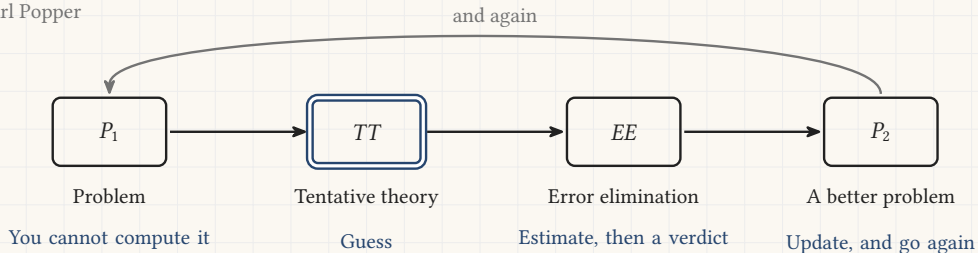
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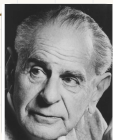
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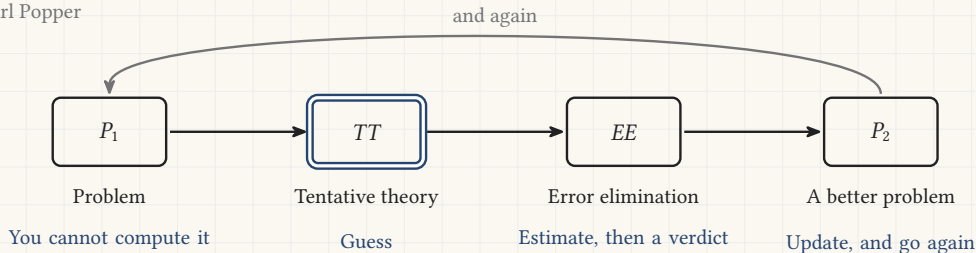
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This weekend is that diagram, and how it produces the AI applications you already use.

Schema after Karl Popper, *Objective Knowledge*, chapter 8 (1972).

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Astronomy: the recovery of Ceres, 1801



Carl Friedrich
Gauss, 1840.

C. A. Jensen

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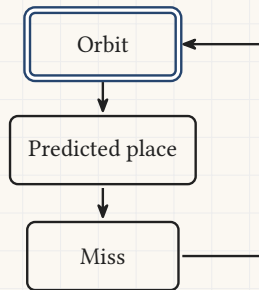
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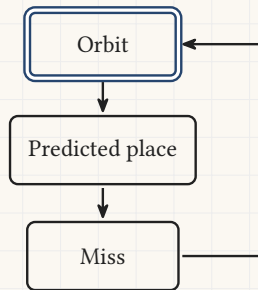
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- Von Zach finds it again in December.



Computer science: Turing, 1950



Alan Turing, 29 March 1951.
Elliott and Fry

A. M. Turing, "Computing Machinery and Intelligence", *Mind* LIX(236), 1950,
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We cannot expect to find a good child-machine at the first attempt. One must experiment with teaching one such machine and see how well it learns. One can then **try another** and see if it is better or worse.

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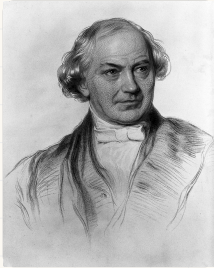
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The loop was written down as the plan for machine intelligence before there was a machine to run it on.

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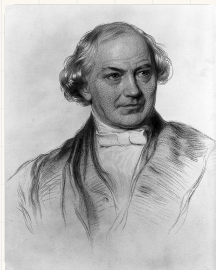
Whewell, 1858: trying wrong guesses



William Whewell, stipple engraving.
Wellcome Collection, CC BY 4.0

William Whewell, *Novum Organon Renovatum*, third edition (1858), book II, chapter V, page 79.

Whewell, 1858: trying wrong guesses

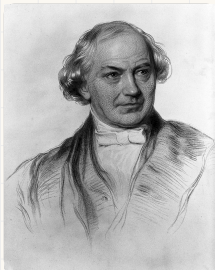


William Whewell, stipple engraving.
Wellcome Collection, CC BY 4.0

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Whewell, 1858: trying wrong guesses



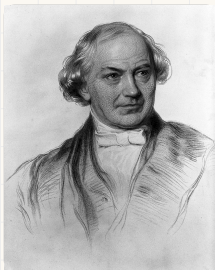
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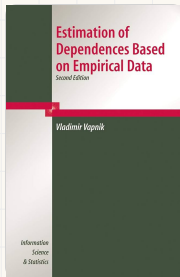
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Most guesses are wrong, and a step that aims too far is how the right one gets found.

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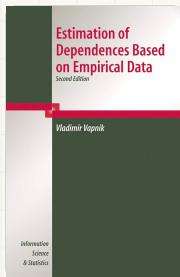
The loop has a name, and the name is a book



Second edition, Springer, 2006

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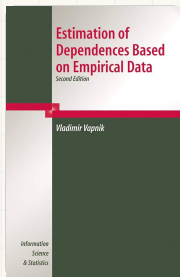


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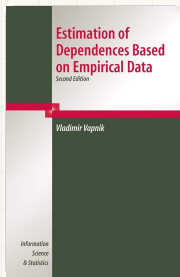
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The book sets out **empirical risk minimisation**, which is the strategy underneath most of what the rest of this course will build.

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- Gradient descent
- Neural networks and universal approximation
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- Validation and overfitting
- The project: routing between language models

Every lecture is followed by an exercise session. Times, rooms and breaks are in the onboarding slides.

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References

Texts

- Popper, Karl R. *The Logic of Scientific Discovery*. London: Hutchinson, 1959. English of *Logik der Forschung*, Vienna: Springer, 1934. Chapter 10, section 85.
- Popper, Karl R. *Conjectures and Refutations*. London: Routledge, 1963. Preface. And *Objective Knowledge*. Oxford: Clarendon, 1972. Chapter 8.
- Turing, A. M. “Computing Machinery and Intelligence”. *Mind* LIX(236), 1950, pages 433 to 460. Section 7.
- Whewell, William. *Novum Organon Renovatum*. Third edition. London: John W. Parker, 1858. Book II, chapter V.
- Vapnik, Vladimir N. *Estimation of Dependences Based on Empirical Data*. Tr. Samuel Kotz. New York: Springer, 1982; 2nd ed. 2006.

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